Haoyang Chen

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Research Interests

I aim to turn World Action Models (WAMs) into robot policies that work in the real world. This requires three coupled properties: consistency over long-horizon tasks through persistent memory and state, inference fast enough for closed-loop control on real robots, and robustness to the variability of real environments and user demands. I approach these through algorithm-system co-design.

Education

University of Minnesota, Twin Cities — *Minneapolis, MN, USA*

B.S. in Computer Science — *May 2026*

- GPA: 3.87, Dean's List

Publications

[1] Shiyang Li*, **Haoyang Chen***, Mattia Fazzini, Caiwen Ding. **CUDABeaver: Benchmarking LLM-Based Automated CUDA Debugging.** *NeurIPS 2026, Evaluations and Datasets Track (Poster).* (*equal contribution) [arXiv] [Code]

[2] **Haoyang Chen**, Caiwen Ding. **CamCanvas: Absolute Camera Placement in Video World Models.** *Under review at ICLR 2027.*

[3] Yunhao Zhao*, Zhenyang Ni*, **Haoyang Chen**, Ruohan Zhang, Qi Zhu. **Skills in Weights, Memory in Code: Hybrid Learning for Memory-Dependent Robot Manipulation.** *Under review at AAAI 2027.* (*equal contribution) [arXiv]

[4] Yuebo Luo, **Haoyang Chen**, et al. **Runtime acceleration for zeroth-order LLM fine-tuning on GPUs.** *Under double-blind review at a systems conference.*

Research Experience

APEX Lab, University of Minnesota, Twin Cities — *Minneapolis, MN, USA*

Research Assistant, Advisor: Prof. Caiwen Ding — *Feb 2026 - Present*

- **Video world models (first author, CamCanvas).** Identified that relative-pose camera interfaces prevent interactive video world models from returning to previously visited viewpoints; proposed a panoramic canvas memory that conditions generation on the rendered target view, raising 90° return success from 0% to 91% and generalizing across architectures.
- **LLM-based CUDA debugging (co-first author, CUDABeaver, NeurIPS 2026).** Co-built a benchmark of 213 real failing CUDA workspaces and the protocol-conditional metric pass@k(M, C, A), exposing "repair by degeneration" where fixes pass tests but abandon the original optimizations.
- **Efficient LLM fine-tuning runtime.** Profiled zeroth-order fine-tuning on GPUs and contributed to an adaptive runtime that removes its memory-layout and execution-configuration bottlenecks (under review).

Northwestern University — *Remote*

Research Collaborator, with Prof. Qi Zhu and Prof. Ruohan Zhang — *Summer 2026*

- **Memory-dependent robot manipulation (HyMeS).** Built the system implementation of a hybrid framework that keeps motor skills in a frozen $\pi_{0.5}$ VLA and learns task memory as executable code refined by a coding agent, steering flow-matching actions with memory-conditioned constraints; raised task success from 41.3% to 60.1% on RoboMemArena and from 25.7% to 57.1% on a real SO-101 arm.

Open Source

vLLM-Omni — *github.com/vllm-project/vllm-omni*

Contributor — *2026 - Present*

- Contributing inference optimizations for World Action Models (WAMs) in the vLLM ecosystem's omni-modality serving framework.

Skills

Programming	Python, C/C++, CUDA, Bash
ML Frameworks	PyTorch, HuggingFace Transformers, vLLM, vLLM-Omni
Systems	GPU profiling and benchmarking, multi-GPU clusters, Linux, Git
Languages	Chinese (Native), English